Activity#3

CODE:

main.py

```python
from ArrayStack import ArrayStack as Stack

# Part 1: Balanced Expression Checker
1 usage  new *
def is_balanced(expression):
    stack = Stack()
    matching_pairs = {')': '(', ']': '[', '}': '{'}

    for char in expression:
        if char in '([{':
            stack.push(char)  # Push opening bracket to the stack
        elif char in ')]}':
            if stack.is_empty() or stack.pop() != matching_pairs[char]:
                return False  # unbalanced if stack top is not a closing bracket

    return stack.is_empty()  # balanced if stack is empty at the end


1 usage  new *
def test_balanced_expressions():
    expressions = [
        "( )(( )){([( )])}",  # Correct
        "((( )(( )){([( )])}))",  # Correct
        ")(( )){([( )])}",  # Incorrect
        "({[])}",  # Incorrect
        "("  # Incorrect
    ]

    print("Balanced Expression Checker Results:")
    for exp in expressions:
        result = is_balanced(exp)
        print(f"Expression: {exp} - Balanced: {result}")
    print("\n")  # space for readability


# Part 2: Reversing Lines in a File
1 usage  new *
def reverse_lines_in_file(input_file, output_file):
    stack = Stack()
```

```
37
38      # Try to open the input file
39      try:
40          with open(input_file, 'r') as file:
41              for line in file:
42                  stack.push(line)
43
44      except FileNotFoundError:
45          print(f"Error: The file {input_file} was not found.")
46          return
47
48      # Open the output file in write mode
49      with open(output_file, 'w') as file:
50          while not stack.is_empty():
51              reversed_line = stack.pop()
52              file.write(reversed_line)
53
54      print("Reversed lines written to {output_file} \n")
55
56

    1 usage  new *
57  def reverse_file_example():
58      input_file = 'myfile.txt'
59      output_file = 'reversed_output.txt'
60
61      # Reverses the lines in the file
62      reverse_lines_in_file(input_file, output_file)
63
64
65  # Run balanced checker and the file reverser
66  if __name__ == "__main__":
67      print("Part 1: Balanced Expression Checker")
68      test_balanced_expressions()
69
70      print("Part 2: Reverse File Lines")
71      reverse_file_example()
```

ArrayStack.py

```
1 usage  new *
class ArrayStack:
    new *
    def __init__(self):
        self.stack = []

    5 usages  new *
    def is_empty(self):
        return len(self.stack) == 0

    2 usages  new *
    def push(self, item):
        self.stack.append(item)

    2 usages  new *
    def pop(self):
        if not self.is_empty():
            return self.stack.pop()
        return None

    new *
    def peek(self):
        if not self.is_empty():
            return self.stack[-1]
        return None

    new *
    def size(self):
        return len(self.stack)
```

myfile.txt

```
this is the first line
this is the second line
this is the third line
this is the fourth line
this is the fifth line
```

OUTPUT:

```
Part 1: Balanced Expression Checker
Balanced Expression Checker Results:
Expression: ( )(( )){([( )])} - Balanced: True
Expression: ((( )(( )){([( )])})) - Balanced: True
Expression: )(( )){([( )])} - Balanced: False
Expression: ({[])} - Balanced: False
Expression: ( - Balanced: False


Part 2: Reverse File Lines
Reversed lines written to {output_file}
```

reversed_output.txt

main.py | reversed_output.txt

```
1  this is the fifth line
2  this is the fourth line
3  this is the third line
4  this is the second line
5  this is the first line
```